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ation (56.8%), overall health (59.5%), self-esteem (100.0%), and occlusion (62.2%). ence of a multidisciplinary healthcare team for managing these patients was noted. ion, it can be inferred that dentists possess a comprehensive understanding of the diverse challenges confronted by patients in the context of prosthetic rehabilitation following oral oncologic surgery. Additionally, the study underscores the pivotal role of prosthetic rehabilitation in enhancing patients' functional abilities, aesthetics, and overall comfort.

Indexing terms: Mouth neoplasms. Radiotherapy. Dental prosthesis.

RESUMO

Objetivo: avaliar se os cirurgiões dentistas compreendem os desafios que os pacientes têm acerca da reabilitação protética pós cirurgia oncológica bucal. **Métodos**: trata-se de um estudo de caráter descritivo-quantitativo. A população foi composta por 37 cirurgiões dentistas de diversas especialidades na cidade de Montes Claros – Minas Gerais, Brazil. A coleta de dados se deu através de um questionário semiestruturado online buscando informações sobre conhecimento e percepção desses profissionais sobre a reabilitação protética em pacientes pós cirurgia oncológica bucal. **Resultados**: nota-se que, 75,7% dos participantes relataram que a maioria dos pacientes se sentiram satisfeitos com o tratamento protético, entre as melhorias, destacam-se, mastigação (94,6%), respiração (35,1%), músculos bucais (48,6%), voz (56,8%), saúde (59,5%), autoestima (100,0%) e mordida (62,2%). Além disso, observou-se a presença de uma equipe multiprofissional para tratamento desse paciente. **Conclusão**: conclui-se que os cirurgiões dentistas compreendem os diversos desafios enfrentados pelos pacientes em relação à reabilitação protética pós cirurgia oncológica bucal e ainda observa-se a importância de que essa reabilitação tem para os pacientes para que haja o desenvolvimento das funções, estética e conforto do paciente.

Termos de indexação: Neoplasias bucais. Radioterapia. Prótese dentária.

INTRODUCTION

Oral cancer may manifest in various clinical presentations. Initially, it often appears asymptomatic, with manifestations including leukoplakic, erythroplakic, and leukoerythroplakic lesions. The primary lesion type is typically characterized by exophytic or endophytic ulcers. In advanced stages, oral cancer can become symptomatic, exhibiting extensive lesions with a firm, nodular mass [1-4].

Head and neck malignancies can arise within the oral cavity, encompassing sites such as the oral mucosa, gingiva, hard palate, tongue, floor of the mouth, oropharynx, nasopharynx, hypopharynx, nasal cavity, paranasal sinuses, larynx, and salivary glands [2].

Tobacco usage stands as a predominant risk factor for oral cancer, with over 70 identified carcinogenic agents, including nitrosamines and polycyclic hydrocarbons (e.g., benzopyrene). Contact of these substances with the oral mucosa can induce thermal injury, promoting chronic inflammation that predisposes to the development of other lesions. While alcoholism alone is not commonly observed as a direct cause of oral carcinomas, its combination with smoking significantly increases the risk, as alcohol dissolves the toxins present in cigarettes [1].

The preferred treatment approach for preventing recurrences and metastases in oral cancer involves heless, radiation doses can lead to adverse effects, particularly on the oral mucosa, mplications such as mucositis, xerostomia, dysgeusia, trismus, radiation-induced oradionecrosis. These effects can significantly impact the patient's quality of life and the overall oncological treatment [6].

Recent years have witnessed substantial progress in systemic therapy for oral cancer, including chemotherapy and immunotherapy. These therapies play an increasingly vital role in managing oral cancer. Chemotherapy has historically been employed as a primary treatment modality or in conjunction with radiation therapy, often serving as a radiation sensitizer. Adjuvant chemotherapy is frequently administered following surgery to reduce the risk of metastatic recurrence [7].

Chemotherapy can detrimentally affect salivary flow, leading to increased cariogenic microorganism activity. Complications may include mucositis, infections, neurotoxicity, and oral pain. Dental caries lesions in these patients typically manifest in the cervical and incisal regions of teeth, progressing rapidly and causing significant discomfort. Radiotherapy-related complications also encompass hyposalivation, characterized by a rapid decline in saliva's buffering capacity. Periodontal issues and osteoradionecrosis, the most severe radiotherapy complication, may also arise [6].

Another approach involves the use of provisional obturator prostheses to address buco-sinus communication, thereby separating the oral and nasal cavities. They aid in the healing process, mitigating the risk of inflammation and infections. These prostheses are applied immediately post-surgery for lesion removal [6,7]. Consequently, there is an indication for employing an obturator prosthesis, whether it be immediate, delayed, or provisional [8]. These prostheses offer an alternative to restore each patient's oral function and aesthetics. However, in certain instances, they may lead to complications related to chewing and swallowing food. Additionally, a variety of lesions may arise in connection with removable prostheses, including stomatitis, traumatic ulcers, periodontal issues, hyperplasia, and candidiasis, with the latter being the most common type of infection [9]. Hence, this study aims to evaluate whether Dental Surgeons comprehend the challenges that patients face in terms of prosthetic rehabilitation post-oral cancer surgery.

METHODS

This study explores the utilization of provisional obturator prostheses as a therapeutic approach for closing buco-sinus communications and facilitating post-operative healing in oral cancer patients. Additionally, it investigates the challenges and complications associated with prosthetic rehabilitation, including their impact on patients' daily lives. A descriptive-quantitative methodology was employed, involving interviews with dentists in the *Montes Claros* region of *Minas Gerais*, Brazil.

Data collection utilized multiple-choice questions administered through the Google Forms® platform. The study, conducted between March and April 2022, included 37 participating dentists from various specialties. The research focused on dentists' communication of indications and contraindications for prosthetic rehabilitation after oncological surgery, patient compliance with prosthesis care, the significance of aesthetics and function in patients' lives, financial resources for treatment, and the efficacy and safety of

RESULTS

A total of thirty-seven Dental Surgeons were interviewed, with 100% of volunteers consenting to participate, as indicated in the Informed Consent Form (ICF). It is noteworthy that the majority of participants were female, accounting for 54.1% of the sample. In terms of graduation years, the highest representation was from the year 2021, followed by 2018, with subsequent years such as 1997, 2007, 2009, 2012, and 2016 having an equal number of graduates. Participant ages ranged from 22 to 65 years. The study encompassed various dental specialties, including general dentistry, prosthodontics, general practice, endodontics, public health in a multidisciplinary context, stomatology, implant dentistry, pediatric dentistry, orthodontics, and family health.

Evaluation of patient satisfaction following prosthetic treatment revealed that 75.7% of respondents reported that most patients expressed satisfaction with the prosthetic intervention. Additionally, 21.6% of respondents indicated that most patients were highly satisfied, while 2.7% reported relatively lower levels of satisfaction (table 1).

Table 1. Dentists' understanding of the challenges that patients have regarding prosthetic rehabilitation after oral cancer surgery in *Montes Claros* (MG).

1 of 2

Variables	%
Degree of patient satisfaction	
Very satisfied	21.6
Satisfied	75.7
Little satisfied	2.7
Dissatisfied	0.0
Aspects Demonstrating Positive Changes	
Chewing	94.6
Breathing	35.1
Buccal muscles	48.6
Voice	56.8
Health	59.5
Self-esteem	100.0

	%
Few changes	0.0
No changes observed	0.0
Discomfort and/or Problems with Oral Prosthesis	
No Discomfort and/or Problem	54.1
No Good Adaptation to Prosthesis	8.1
No Good Adaptation to Prosthesis	13.5
Canker Sores and Ulcers due to Prosthesis	27.0
Prosthesis Causes Pain	8.1
Felt Ashamed for Appearance Change	5.4
No Change Observed	2.7
Average Time to Make Definitive Prosthesis	
1 week	0.0
2 weeks	13.5
3 weeks	54.1
3 months	8.1
3-4 months	2.7
More than 5 months	8.1
Less than a week	5.4
About 2 weeks to make	2.7
Depends on the type of prosthesis	2.7

Source: Research data (2022).

Concerning positive changes observed in patients after oral rehabilitation, the data illustrated improvements in multiple areas: mastication (94.6%), breathing (35.1%), buccal muscle function (48.6%), phonation (56.8%), overall health (59.5%), and self-esteem (100.0%) (table 1).

Analysis of prosthesis usage revealed certain discomforts and challenges encountered in patients' daily lives. Specifically, 54.1% of participants reported no discomfort or issues, 8.1% experienced difficulty adapting to the prosthesis, 13.5% developed canker sores and ulcers attributed to prosthesis use, 27.0% reported discomfort or pain associated with prosthesis use, 8.1% expressed embarrassment due to changes in appearance post-surgery, 5.4% observed no perceptible changes, and 2.7% noted an initial adjustment period for speech adaptation. Ultimately, post-prosthesis fitting, patients generally reported satisfaction (table 1).

Regarding the Timeframe for Fabricating Definitive Prostheses, it was observed that 54.1% of the type of prosthesis, and 2.7% reported an average completion time of around 2

An intriguing aspect regarding the payment arrangements for prostheses provided to patients is that they are commonly covered by the Sistema Único de Saúde (SUS, Unified Health System), private insurance plans, or self-funded by the patients themselves. As depicted in figure 1, the majority of cases (67.6%) indicate that patients bear the full cost of the prosthesis.

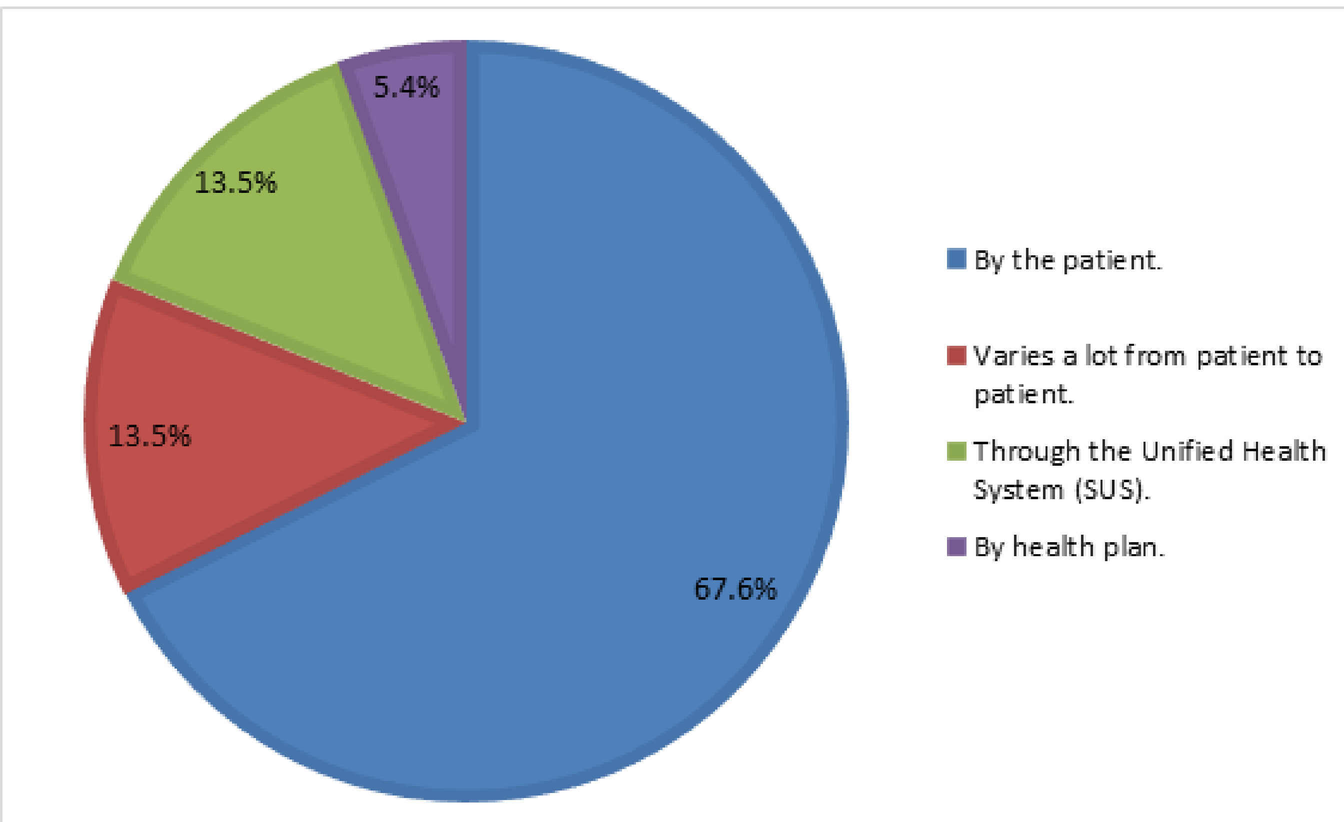

Figure 1. Depicts the payment modalities employed for prosthetic interventions performed on patients in *Montes Claros* (MG), Brazil.

In the query pertaining to the quantity of prostheses required by patients before obtaining the definitive prosthesis, as illustrated in figure 2, it is evident that in 75% of cases, patients needed to utilize only a single prosthesis until the completion of the final one.

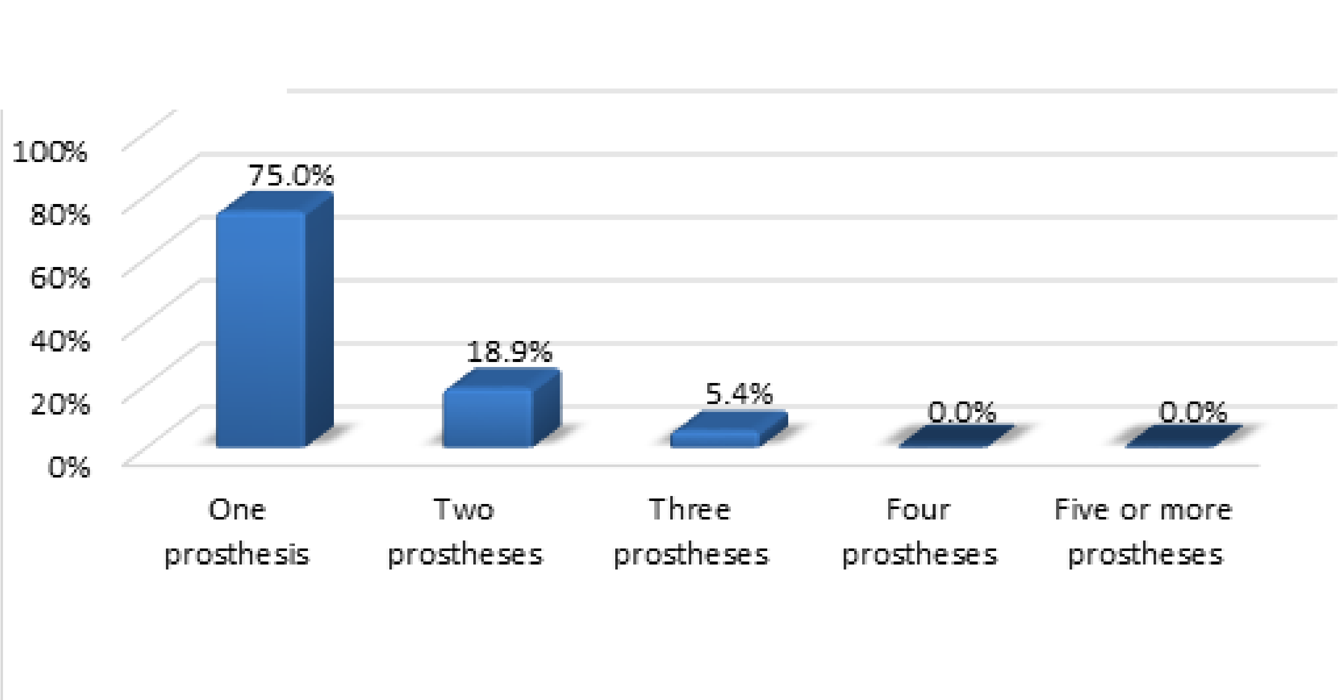

Figure 2. Quantification of prosthetic appliances required by patients in *Montes Claros, Minas Gerais*, to obtain the definitive prosthesis.

Table 2. Health Professionals' Involvement in Assisting Patients During Cancer Treatment in *Montes Claros* (MG), Brazil.

Variables	%
Which health professionals assist patients during cancer treatment?	
Physician	86.5
Dental surgeon	81.1
Maxillofacial surgeon	37.8
Psychologist	59.5
Psychiatrist	13.5
Nutritionist	32.4
Physiotherapist	27.0
Speech therapist	32.4
Nurse	51.4
Nursing technician	27.0
Health agent	24.3
I have never treated a cancer patient	2.7

ygiene habits, and 5.4% experienced a deterioration compared to other patients.

period to the definitive prostheses was predominantly measured in weeks for the .7% – figure 3A). Additionally, following tumor removal through surgical intervention, the results indicate that patients, upon facing the loss of the extracted site, experienced a range of emotions, with 50.0% expressing concerns about aesthetics and 25.0% reporting a sense of relief (figure 3B).

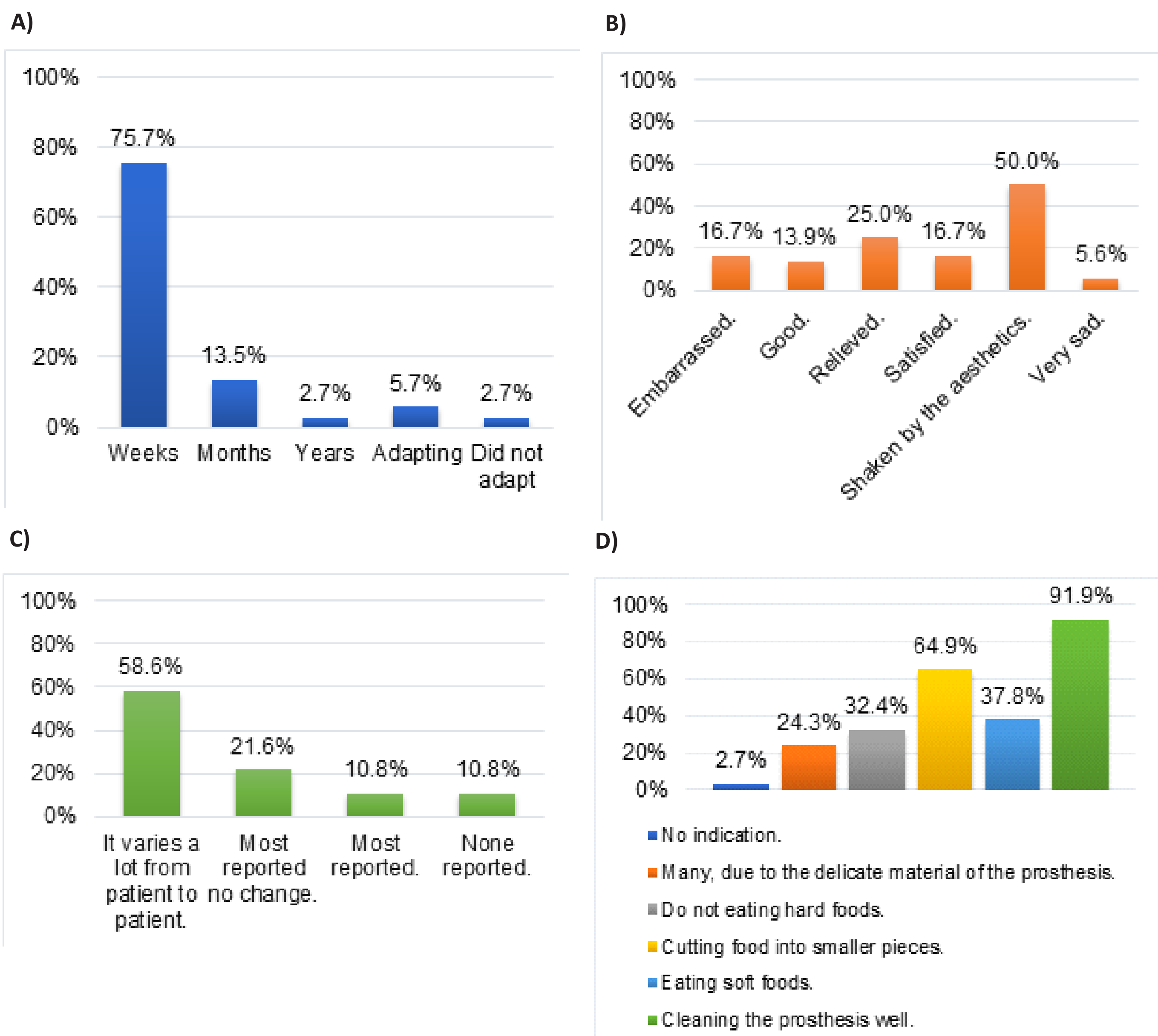

Figure 3. Dentists' responses regarding prosthesis utilization by patients in *Montes Claros*, MG, Brazil: A) How long did it take for patients to adapt to their definitive prostheses? B) After tumor removal through surgery, what was the emotional state of these patients regarding the loss of the extracted site? C) Did patients report the development of lesions or any complications attributed to prosthesis usage? D) What recommendations and contraindications should be provided following oral

nt of lesions and/or complications attributed to prosthesis usage (figure 3C).

about the recommended indications and contraindications following oral e patients, 91.9% of dentists emphasized the importance of thorough prosthesis cleaning, while 64.9% advised patients to cut solid foods into smaller pieces if they were excessively dense (figure 3D).

It is noteworthy that all participants indicated that they consistently instruct patients on proper prosthesis cleaning techniques. In terms of patients' self-esteem and satisfaction levels while using the prosthesis, the majority assigned a score of 8 (32.4%) (table 3).

Table 3. Evaluation of self-esteem and satisfaction levels observed in patients receiving dental prosthesis treatment in *Montes Claros* (MG), Brazil.

Variables	%
On a scale of 0 to 10, what is the extent of self-esteem and satisfaction observed in patients regarding their use of dental prostheses?	
0	0.0
1	0.0
2	0.0
3	8.1
4	10.8
5	16.2
6	18.9
7	10.8
8	32.4
9	2.7
10	0.0

Source: Research data (2022).

Regarding the observation of a complex surgical case involving the removal of one or more tumors, which was extensively studied and involved multiple specialists, Figure 3 illustrates the following: 43.2% of respondents closely monitored the case, noting its challenging anatomical location; 13.5% closely monitored it and found difficulty primarily due to the tumor's size; 10.8% tracked the case but lacked awareness of its intricacies; 8.1% did not acquire any knowledge or involvement in such cases, and 32.4% did not encounter any cases of this nature.

DISCUSSION

The surgical repercussions of treating oral cancer can profoundly affect patients' daily lives, e affected dental structures. In individuals who have undergone tumor resection or ue to oncological treatments, both patients and dentists encounter the challenges ation aimed at achieving optimal aesthetics, retention, and a comfortable occlusal relationship, all while striving to improve the patient's well-being [11].

In the present study, dentists from diverse specialties shared their daily experiences in providing care to patients seeking prosthetic rehabilitation following oral cavity oncological treatments, which often involve surgical procedures. Notably, aesthetics emerged as a significant factor directly influencing patients' self-esteem and overall satisfaction.

The role of aesthetics extends to the psychological well-being of patients, as it pertains to their self-esteem and social interactions. Patients undergoing such treatments may grapple with psychological, familial, or social acceptance issues that can persist after surgery. The restoration of lost function not only addresses physiological needs but also aims to alleviate feelings of anxiety, facilitating the patient's reintegration into their social milieu [3]. This relationship is underscored by the prominence of self-esteem as the primary catalyst for positive changes in the rehabilitation of these patients, as evident in satisfaction levels rated between 8 and 10, with due consideration given to both satisfaction and self-esteem. This priority is followed by factors related to mastication, occlusion, overall health, phonetics, oral muscle function, and respiration (table 1; table 3). A similar trend is observed immediately following oncological surgery, illustrating how aesthetics can significantly impact patients' emotional states in light of the surgery's effects on their facial or oral characteristics. Nonetheless, despite emotional turbulence, post-oral rehabilitation, positive changes become evident, with self-esteem leading the way, followed by improvements in mastication, overall health, speech, oral muscle function, and breathing (table 1).

Achieving adequate retention in mucous-supported prostheses necessitates adherence to specific standards, including precise molding techniques and careful adjustment of vertical dimensions and occlusion patterns. Furthermore, prostheses must restore each patient's unique facial profile, including facial contours, lip support, facial harmony, and functional requirements [3].

The findings of this study underscore the critical importance of achieving a high-quality impression. It is noteworthy that, in most cases, the deadline for obtaining a definitive prosthesis, whether provisional, immediate, or final, falls within a three-week timeframe, accommodating the individualized needs of each patient. Despite the guidelines outlined in the National Oral Health Policy and the provisions of the Brazilian Federal Constitution (Article 196), which stipulates that "Health is a universal right and the responsibility of the State, encompassing access to health promotion, protection, and rehabilitation actions and services" [12], this study reveals that patients bear the entire cost of these prostheses.

This research highlights the necessity for patients to receive immediate prostheses as part of their rehabilitation treatment from the outset. The restoration of lost functions is just as critical as aesthetics in ensuring patient satisfaction and comfort post-rehabilitation. Achieving patient satisfaction requires a multidisciplinary approach, encompassing surgical, psychological, and prosthetic facets, in addition to other functions, and involving multiple healthcare professionals. This collaborative approach contributes to the holistic well-being of patients, addressing both the disease and the emotional and social aspects of their recovery journey. Regular follow-up visits to dental professionals represent a pivotal aspect of ongoing care, erapists, speech therapists, nursing technicians, and health agents are secondary patients. Multidisciplinary approaches and supportive care have been reported to tion and enhance outcomes and quality of life [13], with studies [14,15] indicating a high suicide rate among oral cancer patients due to decreased quality of life.

Regarding discomfort or issues related to prostheses, the majority of patients did not report any such problems. A segment of dentists did not observe any changes, while others noted that some patients experienced pain, canker sores, ulcers, and difficulties with adaptation, all of which could be resolved through adjustments to enhance patient satisfaction (Table 1). Therefore, it is imperative to grasp the principles of proper impression techniques to optimize patient comfort. Impressions are particularly crucial, as they directly influence retention, stability, and support. Furthermore, it is worth noting that bone resorption, alveolar ridge loss, vertical dimension of occlusion reduction, impaired masticatory function, and phonetic alterations can all impact retention [16]. This study revealed that patients typically require weeks to achieve satisfactory adaptation, and the occurrence of injuries due to poorly adapted prostheses along anatomical contours may vary from patient to patient.

Prosthesis replacement hinges on its condition within the oral environment and its structures, as well as patient cooperation. While replacement timing can differ among patients, the study indicated that most patients do not return for routine evaluations.

The research underscores the significance of recommendations provided by dentists to patients. Notably, the most emphasized recommendation was prosthesis cleaning, alongside advice to cut firm foods into smaller pieces to prevent potential damage. It is advisable to avoid overly rigid foods, as these can potentially damage delicate prosthetic materials. Patients are encouraged to start with softer foods and gradually transition to firmer textures. Responsibility for prosthesis cleaning lies with the patient, but it is the dentist's role to motivate and instruct patients on proper hygiene practices. Patients should be aware that prostheses can harbor pathogenic microorganisms, which must be effectively eliminated through gentle brushing with a neutral detergent under running water, followed by thorough drying and safe storage away from children [17]. In this questionnaire, a specific recommendation from one of the research participants was noted, suggesting that patients should "Practice ''s'' sounds in front of the mirror if the initial speech changes are bothering you, among other recommendations." All respondents reported instructing patients on prosthesis cleaning, with most patients subsequently demonstrating improved oral hygiene practices, while a minority maintained their existing habits and a smaller group experienced a deterioration in oral hygiene.

Dentist to make the necessary adjustments. Practice ''s'' sounds in front of the mirror if the changes in speech at the beginning of the adaptation are bothering you. Among other recommendations that can be given, everyone who went through the survey reported informing patients about cleaning the prosthesis. As a result, most of these patients improved their oral hygiene, and a small portion did not change their hygiene habits, with a minority experiencing a decline.

Finally, we have the finding that many professionals have encountered clinical cases with challenging access due to the size of the tumor, necessitating the removal of one or more structures over the course of their careers. The complexity of these tumors and the associated access issues can be aided by various types of imaging exams. However, their intricacies persist due to numerous factors, including potential he study highlights the importance of rehabilitating patients who have undergone e restoration of lost functions is equally significant as addressing aesthetics, patient ehabilitation comfort. It is evident that dentists comprehend the diverse challenges faced by patients in the context of prosthetic rehabilitation following oral cancer surgery.

Collaborators

EN Oliveira, MLA Campos Filho:Preparation of the work, design of the work, and writing the work. MCS Lopes and TAX Santos: Revising and final approval of the work. WVS Pereira: Translator, revising, and final approval of the work. MA Bezerra Junior: Conception of the work, acquisition and analysis of data, and preparation of the work.

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